



UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT4163 - Digital Image Processing
Practical 02 – Basic Operations on Image



- ***Note that: Download the relevant resources for this practical from the resources folder of the LMS.***
1. Load and display the given image “image_02.png” to your Google Colab environment
 2. Write the image back to the Google Colab environment as “Original Image.jpg”
 3. Read and assign the “Original Image.jpg” image to “Org_image” variable on your Google Colab environment.
 4. Display the Original Image using “Org_image” variable on your Google Colab environment.
 5. Modify the pixel values of the given image at the range of [0:9,0:9] to the given color parameters.
 - Color parameters [255, 10, 255]
 - *Note that By default, OpenCV loads images in BGR (Blue, Green, Red) format, while most other libraries (e.g., Matplotlib, PIL) use RGB (Red, Green, Blue) format.*
 - Observe the difference between the original image and the Image after modifying pixel values at the range of [0:9,0:9]:
 6. Access and Display the following properties of the given image.
 - a) Height,
 - b) Width
 - c) Number of channels
 - d) Total number of pixels
 - e) Data type
 7. Extract an ROI of 100x100 pixels from the top-left corner of the given image and write the new image as “ROI.jpg” in to the Google Colab environment.
 8. Load the “opencv_logo.png” image into your Google Collab environment which is given in the resource folder and split that image into its color channels (B, G, R) and display them separately.

9. Merge the separated color channels into a single image as “Merged_opencv_logo” and write it back into the Google Colab as “Merged_opencv_logo.png”.
10. Declare two variables as MergedOpenCvLogo and OriginalOpenCvLogo, and read and display the “Merged_opencv_logo.jpg” and “opencv_logo.png” respectively through the declared variables, and observe is there any difference between the displayed images.
11. Read the image “opencv_logo.png” and split in into the separate color channels as B,G,and R and Merge and Swap the color channels to obtain the following output.



Figure 1 - Original Image

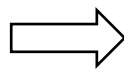


Figure 2- Color Channel swapped Image (Expected Output)

12. Find two suitable images on the internet and perform the following Arithmetic operation on that, and write your final image to the Google Colab. *(Use two separate set of images for each operation and apply your creativity for this task)*
 - I. Addition
 - II. Subtraction
13. Find two suitable images on the internet and perform the blending operation on that, and write your final image to the Google Colab. . *(Use your creativity for this task and apply suitable values for α , β and γ .)*
14.
 - a) Draw a rectangle image as a 300 x 300 NumPy array.
 - b) Draw a white filled rectangle on the black image using CV2. *(Use the (25, 25) as the Top-left corner and the (275, 275) as the Bottom-right corner of the rectangle.)*
 - c) Draw a rectangle image as a 300 x 300 NumPy array.
 - d) Draw a filled white circle on the black image according to the following properties.
 - The center of the circle : $x=150, y=150$
 - The radius of the circle: 150 pixels).
 - Color: White
 - e) Perform the Bitwise Operations (AND, OR, XOR, NOT) using the drawn images above ‘b’ and ‘d’.